

Main.jsx code explained:

This is the **entry point of our entire React application**.
Think of it as the “**starting engine**” of our facial recognition frontend.

1. IMPORTS (LOADING WHAT WE NEED)

```
import React from 'react';  
import ReactDOM from 'react-dom/client';  
import './index.css';  
import App from './App';
```

What each line does:

◇ `import React from 'react';`

Loads the React library

- Required to use JSX (`<App />`)
- Core engine for building UI

◇ `import ReactDOM from 'react-dom/client';`

This connects React to the **browser (DOM)**

- React builds UI in memory
- ReactDOM **renders it on the actual web page**

◇ `import './index.css';`

Loads our **global styles**

- Applies:
 - layout
 - fonts
 - colors
 - centering

✓ This affects the entire app

◇ `import App from './App';`

Imports our **main component**

- This is our:
 - upload form
 - API logic
 - UI

✓ This is our **facial recognition app itself**

2. CREATE ROOT (VERY IMPORTANT)

```
const root = ReactDOM.createRoot(document.getElementById('root'));
```

What's happening:

React is attaching itself to this HTML element:

```
<div id="root"></div>
```

Breakdown:

- `document.getElementById('root')` → finds the HTML container
- `createRoot(...)` → creates a React root (React 18 feature)

Why this matters:

This is where our entire app will live

✓ Without this → nothing shows on screen

3. RENDER THE APP

```
root.render(  
  <React.StrictMode>  
    <App />  
  </React.StrictMode>  
);
```

◇ `<App />`

This loads our main component

- Everything we built:
 - form
 - upload
 - API calls
 - image preview

✓ It becomes visible on screen

◇ `<React.StrictMode>`

Development-only tool

What it does:

- Detects bugs
- Warns about bad practices
- Runs extra checks

FULL FLOW (HOW APP STARTS)

Browser loads `index.html`

↓

`<div id="root"></div>`

↓

`main.jsx` attaches React to root

↓

`<App />` is rendered

↓

Our facial recognition UI appears

BIG PICTURE (VERY IMPORTANT)

This file does ONLY ONE thing:

“Mount the React app into the browser”

REAL ENGINEER ANALOGY

Think of it like:

Part	Meaning
index.html	Empty building
main.jsx	Construction team
App.jsx	Interior design

“This file serves as the entry point of the React application. It initialises the React root using ReactDOM, attaches it to the DOM element with id ‘root’, and renders the main App component. It also imports global styles and optionally uses React StrictMode for development checks.”